Akoizo Original Practice - Form 1 - Answer key

Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) What is the minimum airplane mass excluding the motor?

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

A. 8.0 g

B. 1.5 g

C. 3.5 g

D. 10.0 g

Correct: A

2. (2 points) Why is a stiff motor stick important?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

A. It reduces motor mass

B. It shortens the flight

C. Bending changes thrust direction and trim as torque changes

D. It adds lift

Correct: C

3. (2 points) Why are large, slow propellers efficient for indoor planes?

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

A. They make more noise

B. They spin faster

C. They push a large mass of air gently, wasting less energy

D. They weigh less

Correct: C

4. (2 points) Which drag type is largest at low speed?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

A. Skin friction only

B. Wave drag

C. Induced drag

D. Parasite drag

Correct: C

5. (2 points) Why bank a safe official flight early?

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

A. It guarantees a score before trying a riskier winding

B. Official flights get bonus time

C. The first flight counts double

D. It is required

Correct: A

6. (4 points) Why test glide before powered flights?

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

answer: Glide shows pitch trim without motor effects, so problems like stalls or dives can be fixed safely. (4 points).

7. (4 points) Why is the motor mass limit the main energy limit?

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

answer: Stored energy depends mostly on rubber mass, so a 1.5 g cap limits the total energy available. (4 points).

8. (4 points) Why does the 8.0 g minimum change design priorities?

answer: Teams cannot win by building ultralight, so mass should go into stiffness, durability, and wing area within the box limits. (4 points).

9. (4 points) Why test box fit with the exact flight setup?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

answer: The plane must fit in flight configuration without changing shape; different props or gear can make it too big. (4 points).

10. (4 points) Why is a freewheel allowed but variable pitch not?

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

answer: A freewheel only lets the propeller spin freely; variable pitch actively changes blade angle, which the rules prohibit. (4 points).

11. (6 points) Solve this case and explain your reasoning: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).

reasoning: Weight W = 0.0095 kg x 9.8 m/s^2 about 0.093 N.
v = sqrt (2W ÷ (ρSC_L)) = sqrt (0.186 ÷ (1.2 x 0.040 x 0.8)) = sqrt (4.84).
v about 2.2 m/s. (3 points).

conclusion: The plane cruises near 2.2 m/s; more area or C_L would slow it further. (3 points).

12. (6 points) Solve this case and explain your reasoning: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.

reasoning: Flight time peaks near 0.13 torque units.
At 0.16, the plane hits the ceiling and loses time.
Wind near 0.13 for the first official flight; adjust slightly after watching it. (3 points).

conclusion: Use about 0.13 torque units for a safe first official flight. (3 points).

13. (2 points) What is the likely cause?

Supplied synthetic observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

A. Torque roll during the burst is too strong for the trim

B. CG too far forward

C. Not enough motor turns

Correct: A

14. (2 points) Define “Rubber motor” in the context of Wright Stuff.

answer: Twisted loop of rubber that stores elastic energy. (2 points).

15. (2 points) Define “Airframe mass” in the context of Wright Stuff.

answer: Plane mass without motor; must be at least 8.0 g. (2 points).

16. (2 points) Define “Rib” in the context of Wright Stuff.

answer: Crosswise piece that sets the airfoil shape. (2 points).

17. (2 points) Define “Pitch” in the context of Wright Stuff.

answer: Ideal forward advance per revolution. (2 points).

18. (2 points) Define “Camber” in the context of Wright Stuff.

answer: Curvature of the wing section. (2 points).